

Cloud Security Project Report

AWS EC2 Security Hardening & Monitoring Lab

Executive Summary

This project demonstrates a practical cloud security implementation using AWS EC2, Linux hardening, SSH security configuration, Security Group firewall controls, and AWS CloudWatch monitoring.

Project Objectives

Deploy secure AWS infrastructure, implement SSH key authentication, apply Linux hardening, configure monitoring and document security controls.

Environment & Technologies

AWS EC2, Amazon Linux 2023, Security Groups, SSH, Linux Administration, CloudWatch, SNS.

Implementation Summary

Launched EC2 (t3.micro, Mumbai region), configured Security Group with SSH restricted to trusted IP, connected via SSH key authentication, created cloudadmin user, configured sudo privileges, hardened sshd_config, deployed CloudWatch CPU alarm with SNS notifications.

Security Controls Implemented

Identity & Access: SSH key auth, least privilege, sudo control. Network: Security Group restriction to My IP. Host: root login disabled, password auth disabled, MaxAuthTries configured. Monitoring: CloudWatch alarms and notifications.

Commands Used

sudo dnf update -y | sudo adduser cloudadmin | sudo usermod -aG wheel cloudadmin | sudo systemctl restart sshd

Lessons Learned

Hands■on AWS infrastructure deployment, Linux hardening, SSH configuration, cloud monitoring, security best practices, and operational documentation.

Future Improvements

IAM roles, CloudTrail logging, GuardDuty integration, AWS Config compliance checks, Terraform automation, vulnerability scanning.

Security Controls Matrix

Category	Implemented Control
Identity & Access	SSH Keys, cloudadmin User, Least Privilege
Network Security	Security Group: SSH Port 22 -> My IP Only
Host Security	Disable Root Login, Disable Password Login
Monitoring	CloudWatch CPU Alarm + SNS Email Alerts